

CURRICULUM VITAE – GEORGIOS KISSAS

PERSONAL INFORMATION

Georgios Kissas
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PROFESSIONAL OBJECTIVE

Researcher with extensive experience at the intersection of deep learning, applied mathematics, and science. Proven track record in developing innovative machine learning algorithms, ranging from large-scale simulations of physical systems to scientific discovery based on language modeling. Experienced in leading interdisciplinary research projects, mentoring junior researchers, and driving transformative innovations in data-driven science. Passionate about applying machine learning techniques to solve real-world challenges, with a particular interest in large scale AI for high impact societal applications.

AWARDS

- **AWS Free Credit** Trainium TPU Servers, 2025
- **Top Reviewer** NeurIPS, 2024
- **Asuera Stiftung Scholarship** ETH Zürich Foundation, 120000 CHF, 2023
- **Post Doctoral Fellowship** ETH AI Center (1.5% acceptance rate), 2023
- **Outstanding TA Award** MEAM UPenn, 2021
- **Gerondellis Foundation** Scholarship Award, 2020
- **MIT Travel Grant** Learning for Dynamical Systems and Control, 2019
- **Doctoral Fellowship Award** University of Pennsylvania, 2018
- **Limmat Stiftung Award** Award for Academic Excellence, 2017

PROFESSIONAL EXPERIENCE

- Senior Data Scientist, Research** Swiss Data Science Center *November 2025-Present*
- Working in the area of Large Scale AI for Exascale Cosmological Astrophysics simulations, and Subsonic Turbulence.
- Post Doctoral Researcher & Lecturer** ETH Zurich *June 2025-November 2025*
- Joint Post Doctoral positions between the Seminar for Applied Mathematics, the Departments of Civil, Environmental and Geomatic Engineering, and Mechanical and Process Engineering.
 - Lecturer in the Department of Computer Science
 - Working in the area of deep learning algorithms for scientific discovery through first-principles AI with applications in dynamical systems and cardiovascular biomechanics.
- Post Doctoral Fellow** ETH AI Center *June 2023-June 2025*
- Joint Post Doctoral Fellowship between the Department of Computer Science, the Seminar for Applied Mathematics, the Departments of Civil, Environmental and Geomatic Engineering, and Mechanical and Process Engineering.
 - Working in the area of deep learning algorithms for scientific discovery through first-principles AI with applications in dynamical systems and cardiovascular biomechanics.
- Research Assistant** University of Pennsylvania, PI Lab *August 2018-May 2023*
- Worked in the areas of Deep Learning for physics simulations and precision medicine, Operator Learning, Fluid Dynamics for Cardiovascular Flows, Bayesian Inference, Deep Learning for Dynamical Systems and Sensitivity Analysis.

EDUCATION

- University of Pennsylvania**, Philadelphia, PA, USA
- PhD, Mechanical Engineering and Applied Mechanics, August 2018 - May 2023
Thesis: *"Towards digital twins for cardiovascular flows: A Hybrid Machine Learning and Computational Fluid Dynamics Approach"*
Advisor: Paris Perdikaris
- National Technical University of Athens**, Athens, Greece
- M.S., Computational Mechanics (Fluids), May, 2017, (2nd Highest GPA)
Thesis: *"A computational study of Self-Consistent Field Theory for polymer interfaces"*
Advisor: Doros Theodorou
- University of Patras**, Patras, Greece
- Diploma, Mechanical and Aeronautical Engineering, February, 2015 (Honors)

PATENTS

Computer Systems and Methods for Learning Operators, P. Perdikaris, G.J. Pappas, J. Seidman, **G. Kissas** - Provisional Patent Application 22-9935 (<https://www.freepatentsonline.com/y2023/0214661.html>).

PUBLICATIONS

- Kissas, G.**, 2025, A Principled Substitute for Softmax Attention. (*To appear in the ArXiv*)
- Brivio, S, **Kissas, G.**, Manzoni, A., and Mishra, S., 2025, Reduced-Order Operator Learning through Perceivers: Dissecting the Hidden Mechanisms. (*To appear in the ArXiv*)
- Yu, K., Chatzi, E. and **Kissas, G.**, 2025. Grammar-based Ordinary Differential Equation Discovery. *Mechanical Systems and Signal Processing*, 240, p.113395.
- Oikonomou, O., Lingsch, L., Grund, D., Mishra, S. and **Kissas, G.**, 2025. Neuro-Symbolic AI for Analytical Solutions of Differential Equations. arXiv preprint arXiv:2502.01476.(*Under Review*)
- Renner, D. Mishra, S., and **Kissas, G.**, 2024. Accelerated Patient-Specific Calibration via Differentiable Hemodynamics Simulations. arXiv preprint arXiv:2412.14572.(*Under Review ACM TOMS*)
- Lingsch, L.E., Grund, D., Mishra, S. and **Kissas, G.**, 2024. FUSE: Fast Unified Simulation and Estimation for PDEs, *Thirty-eighth Conference on Neural Information Processing Systems*, 2024.
- Sokolow, R., Bhattacharya, A., **Kissas, G.**, Thompson, E.W., Beeche, C., Swago, S., Zhang, D., Viswanadha, M., Morse, C., Chirinos, J. and Damrauer, S., 2025. Reduced order computational fluid dynamic simulations in the thoracic aorta are associated with disease recorded in a medical biobank. *Scientific Reports*, 15(1), p.39203.
- Kissas, G.**, Mishra, S., Chatzi, E. and De Lorenzis, L., 2024. The Language of Hyperelastic Materials. *Computer Methods in Applied Mechanics and Engineering*, 428, p.117053.
- Seidman, J.H., **Kissas, G.**, Pappas, G.J. and Perdikaris, P., 2023. Variational Autoencoding Neural Operators, *International Conference on Machine Learning*, 2023.
- Kissas, G.***, Seidman *, J.H., Guilhoto, L.F., Preciado, V.M., Pappas, G.J. and Perdikaris, P., 2022. Learning Operators with Coupled Attention. *Thirty-sixth Conference on Neural Information Processing Systems, Journal to Conference Track (Spotlight)*
- Seidman, J.H.*, **Kissas, G. ***, Perdikaris, P. and Pappas, G.J., 2022. NOMAD: Nonlinear Manifold Decoders for Operator Learning, *Thirty-sixth Conference on Neural Information Processing Systems. (Featured in the "Paper of the Week" section of DeepAI)*
- Kissas, G.***, Hwuang, E., Thompson, E.W., Schwartz, N., Detre, J.A., Witschey, W.R. and Perdikaris, P., 2022. Feasibility of Vascular Parameter Estimation for Assessing Hypertensive Pregnancy Disorders. *Journal of Biomechanical Engineering*.
- Yang, Y., **Kissas, G.**, Perdikaris, P., 2022. Scalable uncertainty quantification for deep operator networks using randomized priors. *Computer Methods in Applied Mechanics and Engineering*, 399, p.115399.
- Kissas, G.***, Seidman *, J.H., Guilhoto, L.F., Preciado, V.M., Pappas, G.J. and Perdikaris, P., 2022. Learning Operators with Coupled Attention. *Journal of Machine Learning Research*, 23(215), pp.1-63
- Kissas, G.**, Yang, Y., Hwuang, E., Witschey, W.R., Detre, J.A. and Perdikaris, P., 2020. Machine learning in cardiovascular flows modeling: Predicting arterial blood pressure from non-invasive 4D flow MRI data using physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 358, p.112623.(*Featured in the "Most Cited Articles" section of CNAME as one of the most cited papers of the years 2019-2023.*)

* indicates equal contribution

COMPUTER SKILLS

- ML Frameworks: Pytorch (Distributed, NVIDIA Nsight, NCCL), Jax.
- Applications: Paraview, SimVascular, Mathematica.